4. Program D: 16-bit multiplication and division

Given: multiplication : 1357H X 0025H;

Division : 9A23H ÷ 0011H

Multiplication:

1359H x0025H = 0002 CA13H

Upper word = 0002H

Lower word = CA13H

Division:

9A23H ÷ 0011H

Quotient = 0911H

Remainder = 0002H.

Manual - Calculation

DEVIPRIYA S
[redacted]

Exp - 01

1) Program A: 8-bit addition & Subtraction

given A = 5AH , B = 27H

Addition:

5AH + 27H = 81H

Sum = 81H, Cy = 0

AC = 1, P = 1, Z = 0, S = 1

Flag byte = 94H

Signed overflow = Yes

Subtraction

5AH - 27H = 33H

Diff = 33H, Borrow = 0

AC = 0, P = 1, Z = 0, S = 0

Flag byte = 06H

Signed overflow = No

2) Program B: 16-bit addition and subtraction

given X = 1234H, Y = 4321H

Addition:

1234H + 4321H = 5555H

carry = 0

Subtraction:

1234H - 4321 = CF13H

Borrow = 1

3) Program C: 8-bit multiplication

Multiplication: 14H x 06H; Division F2H ÷ 0AH

Multiplication

14H x 06H = 0078H

Division

F2H ÷ 0AH

Quotient = 18H